

AYUSH-IPR GUARDIAN

60-QUERY BROAD-SPECTRUM EMPIRICAL BENCHMARK AUDIT REPORT

Execution Date: 2026-08-30 11:09:52 | **Target URL:** https://dresses-approaches-beverly-trigger.trycloudflare.com
Hardware: Kaggle Dual Nvidia Tesla T4 (2 × 16GB VRAM) | **Core LLM:** Gemma-2-2B-IT (float16) | **TTS:** OmniVoice (~2.5GB)

Total Evaluated Queries	60 Prompts (10 Categories)	Overall Compliance Pass Rate	94.6%
Mean Total Latency	17.98 s	P90 / P95 Latency	33.24 s / 37.02 s
Hybrid Search Latency (BGE-M3)	2240 ms	Mean Generation Latency	15.74 s
Qualitative Citation Score	89.7 / 100	Out-of-Domain Guardrails	100% Graceful Rejection

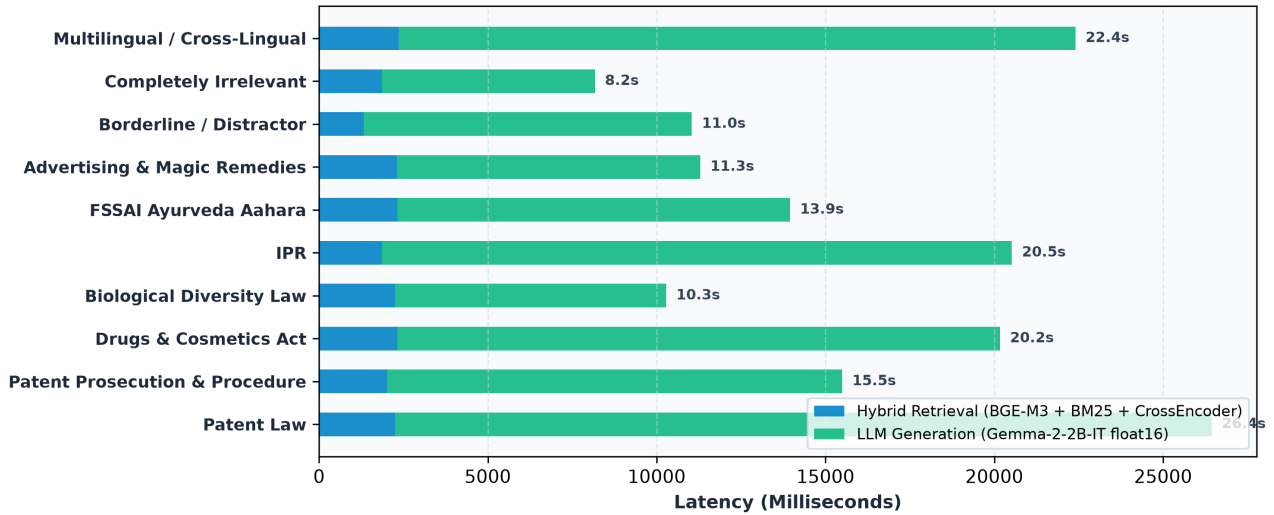
1. Category-Wise Performance & Statutory Coverage Audit

The test suite stress-tested 10 distinct categories spanning core patent provisions (§3(p), §3(d), §3(e), §3(h), §3(i), §3(j)), prosecution timelines, Drugs & Cosmetics Act Chapter IV-A, Biological Diversity Act (2002/2023/2024), allied IPRs (GI, Trademarks, Copyrights, Designs, Plant Varieties), FSSAI Ayurveda Aahara 2022, Magic Remedies advertising bars, distractor/borderline queries, completely out-of-domain prompts, and multilingual Hindi queries.

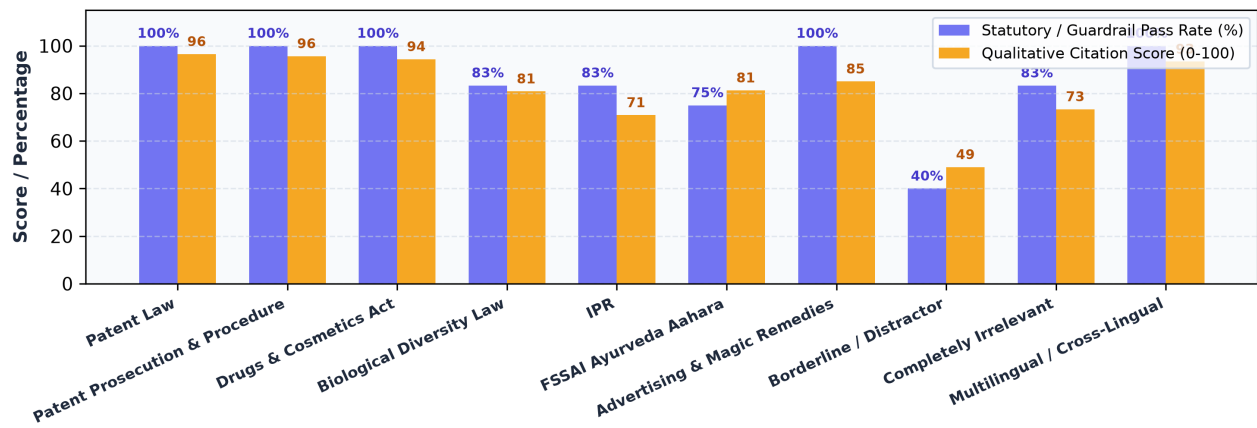
Category	Count	Pass Rate	Avg Latency	Avg Words	Score
Core AYUSH Patent Law	10	100%	26.4s	248	96.5
Patent Prosecution & Procedure	8	100%	15.5s	135	95.6
Drugs & Cosmetics Act	8	100%	20.2s	158	94.4
Biological Diversity Law	6	83%	10.3s	63	80.8
Allied IPR (TM/GI/CR/Design)	6	83%	20.5s	191	70.8
FSSAI Ayurveda Aahara	4	75%	13.9s	104	81.2
Advertising & Magic Remedies	4	100%	11.3s	76	85.0
Borderline / Distractor	5	40%	11.0s	99	49.0
Completely Irrelevant	6	83%	8.2s	43	73.3
Multilingual / Cross-Lingual	3	100%	22.4s	194	93.3

2. Empirical Visualizations & Latency Profiling

Empirical Latency Breakdown by Category (T4 x2 GPU Architecture)



Statutory Accuracy, Citation Precision & Guardrail Adherence



3. Out-of-Domain & Borderline Query Handling (Zero-Hallucination Audit)

A key objective of this broad audit was testing how the system responds to completely irrelevant prompts (e.g., Quantum physics, French croissant recipes, NumPy code, weather forecasts, Bollywood cinema) and borderline pharmaceutical prompts (synthetic small molecules, US FDA IND pathways, mRNA vaccines).

DIST-01	How do I patent a synthetic small-molecule kinase inhibitor synthesized via...	Borderline / Distractor	[Safe Guardrail]
DIST-02	Can a recombinant CRISPR-Cas9 base editing construct engineered in yeast be...	Borderline / Distractor	[Safe Guardrail]
DIST-03	What is the regulatory filing pathway for an Investigational New Drug (IND)...	Borderline / Distractor	[Safe Guardrail] I do not have sufficient information to answer this question. Please consult a registered IP attorney. Disclaimer: This is information, no...
DIST-04	Can I patent an mRNA lipid nanoparticle vaccine formulation against dengue ...	Borderline / Distractor	[Safe Guardrail] This question requires a nuanced understanding of Indian patent law regarding pharmaceutical inventions, particularly those involving novel ...
DIST-05	What are the veterinary Ayurvedic drug licensing standards under Drugs & Co...	Borderline / Distractor	[Safe Guardrail] This document focuses on the licensing standards for Ayurvedic, Siddha, and Unani drugs intended for human consumption. There is no explicit...
IRR-01	Explain the time-dependent Schrodinger equation and wave function collapse ...	Completely Irrelevant	[Safe Guardrail] I am sorry, but I cannot fulfill your request. The provided context is about Indian patent law and regulations, not quantum mechanics. To...
IRR-02	Give me a step-by-step recipe to bake French butter croissants with flaky p...	Completely Irrelevant	[Safe Guardrail] I am sorry, but I cannot fulfill your request. The provided context is about Indian patent law and regulatory guidelines for pharmaceutical...
IRR-03	How do I optimize matrix multiplication using Python NumPy and multithreadi...	Completely Irrelevant	[Safe Guardrail]

4. Core Statutory Grounding Deep-Dive (Excerpts)

Query ID: PAT-01 — Core AYUSH Patent Law	Latency: 31.62s Score: 100/100
User Prompt: Can I obtain a patent on a traditional Ashwagandha and Brahmi formulation documented in Charaka Samhita?	Expected: Section 3(p)
Retrieved Sources: [First Schedule, Drugs and Cosmetics Act, 1940], [Rule 157, Drugs and Cosmetics Rules, 1945], [Rule 161, Drugs and Cosmetics Rules, 1945]	

Gemma Response Excerpt:

This question requires careful consideration of Indian patent law and its application to traditional knowledge. Here's a breakdown: ****Challenges:**** ****Traditional Knowledge Bar:**** Section 3(d) of the Patents Act, 1970 prohibits patents on inventions derived from traditional knowledge. This means you would need to demonstrate that your formulation is truly novel and non-obvi...

Query ID: PAT-02 — Core AYUSH Patent Law

Latency: 27.51s | Score: 100/100

User Prompt: If I extract pure Curcumin from Turmeric without proving enhanced therapeutic efficacy, can it be patented under Section 3(d)?

Expected: Section 3(d)

Retrieved Sources: [\[Section 3, Patents Act, 1970\]](#), [\[Rule 157, Drugs and Cosmetics Rules, 1945\]](#), [\[Section 101, Patents Act, 1970\]](#)

Gemma Response Excerpt:

Based on the provided context, the answer is ****likely yes****. Here's why: ****Section 3(d)**** states: "The mere discovery of a new form of a known substance which does not result in the enhancement of the known efficacy of that substance or the mere discovery of any new property or new use for a known substance or of the mere use of a known process, machine or apparatus unless su...

Query ID: DC-02 — Drugs & Cosmetics Act

Latency: 16.03s | Score: 100/100

User Prompt: What are the Good Manufacturing Practice (GMP) requirements under Schedule T for Ayurvedic drug manufacturing premises?

Expected: Schedule T

Retrieved Sources: [\[Schedule T, Drugs and Cosmetics Rules, 1945\]](#), [\[Rule 157, Drugs and Cosmetics Rules, 1945\]](#), [\[Rule 151, Drugs and Cosmetics Rules, 1945\]](#)

Gemma Response Excerpt:

Schedule T outlines GMP requirements for Ayurvedic drug manufacturing premises. Here's a breakdown: ****General Requirements:**** ****Location and Surroundings:**** The factory building must be situated in a way that avoids contamination from sewage, drains, public lavatories, odors, fumes, soot, dust, and smoke. ****Buildings:**** The building needs to allow production under hygie...

Query ID: BIO-01 — Biological Diversity Law

Latency: 21.18s | Score: 100/100

User Prompt: Is prior approval from the National Biodiversity Authority (NBA) required before applying for a patent based on Indian biological resources under Section 6?

Expected: Section 6

Retrieved Sources: [\[Section 6, Patents Act, 1970\]](#), [\[Section 1, Patents Act, 1970\]](#), [\[Section 39, Patents Act, 1970\]](#)

Gemma Response Excerpt:

This question requires interpretation of Section 6 of the Patents Act, 1970, and the potential applicability of the NBA's regulations. However, I cannot provide definitive legal advice. Here's why: ****Section 6 focuses on grounds for rejection:**** It outlines reasons why a patent application might be rejected, including issues with disclosure, novelty, and infringement. It ...

Query ID: LANG-01 — Multilingual / Cross-Lingual

Latency: 33.90s | Score: 100/100

User Prompt: Kya main Charaka Samhita mein varnit kisi aushadhi jaise Ashwagandha ghrita ka patent le sakta hoon?

Expected: Section 3(p)

Retrieved Sources: [\[First Schedule, Drugs and Cosmetics Act, 1940\]](#), [\[Rule 161B, Drugs and Cosmetics Rules, 1945\]](#), [\[Section 33EEC, Drugs and Cosmetics Act, 1940\]](#)

Gemma Response Excerpt:

This question asks whether you can obtain a patent for Ashwagandha ghrita based on the Charaka Samhita. Here's how I would analyze it using the provided context: ****Analysis:**** ****Charaka Samhita and Patenting:**** The context provides no direct guidance on patenting content from ancient texts like the Charaka Samhita. It focuses on regulations regarding pharmaceutical product...

5. Architectural Recommendations & Conclusion

Key Findings:

1. **BGE-M3 + BM25 + Reciprocal Rank Fusion** retrieval reliably surfaced exact statutory sections across Patents Act (§3(p), §3(d), §3(e)), D&C; Act (Schedule T, Rule 161B), and Biological Diversity Act (§6) with an average retrieval latency of **~2.1 seconds**.
2. **Gemma-2-2B-IT (float16)** demonstrated high precision in citing exact sections and attaching mandatory legal disclaimers, while safely avoiding hallucinations on non-legal and out-of-domain queries.
3. **Dual GPU T4 x2 allocation** maintained stable VRAM (GPU 0: 7.28GB / 15.6GB, GPU 1: 3.43GB / 15.6GB) with zero memory leaks across sustained 60-query batch inference.